

```
theory Week_1_sol
imports Main
begin
```

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Term 1

Computer-Assisted Theorem Proving: 7CCSACTP

Exercise Sheet 1

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called `Week_1.thy` and add the following to its beginning.

```
theory Week_1
imports Main
begin
```

Exercise 1.1 Trying out Isabelle

In this sheet, you should try out the Isabelle theorem prover. Prove the following theorem, first on pen-and-paper, then formally in Isabelle.

lemma

```
assumes T: T b
  and A: A a  $\wedge$  A b  $\implies$  a = b
  and TA:  $\bigwedge x. T x \implies A x$ 
  and Aab: A a
shows T a
```

proof—

— *proof* — starts a structured proof without applying any method, leaving the goal unchanged.
— Style: declarative (Isar *have/show*) with apply-style steps; forward (*OF*); backward (*rule*); *subst*.

have 1: A b

— *have* states an intermediate fact (here named *1*) and proves it.

```

using  $TA[OF\ T]$ 
— using passes facts to the proof method that follows.
.
— . proves the goal directly by a fact given with using, instantiating its variables if needed.

have 2:  $A\ a \wedge A\ b$ 
apply(rule conjI)
— rule r matches the conclusion of r with the goal and replaces the goal by r's premises
(backward reasoning); here conjI leaves  $A\ a$  and  $A\ b$ .
using  $Aab\ 1$ 
.

have 3:  $a = b$ 
apply(rule A)
using 2
.

show ?thesis
— show proves the goal of the enclosing proof; ?thesis abbreviates that goal.
using  $T$ 
apply(subst 3)
— subst eq rewrites the goal with the equation eq, left to right; here a becomes b.
.
qed
— qed closes the proof block, once show has proved its goal.

end

```